THE LOWER SHRIEK FUNCTORS COMPOSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G55 (sources/stacks-duality/pinned-original-source.tex line 8882).

Allowed standard prerequisites (editorial).

- Duality for compactly supported pushforward. Same chapter, Proposition labelled proposition-duality-compactly-supported, cited at proof lines 8933-8945. Relates the lower shriek pushforwards to the upper shrieks.
- Composition of the upper shriek functors. Same chapter, Lemma labelled lemma-upper-shriek-composition, cited at proof lines 8939-8941. Gives f^!g^! = (g o f)^!.
- The pro-category description of morphisms of lower shriek objects. Categories, Remark labelled categories-remark-pro-category-copresheaves, cited at proof lines 8922-8924, and Same chapter, Remark labelled remark-composition-lower-shriek, invoked in the statement. Reduces the claim to testing Hom into bounded coherent objects.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G54: sources/stacks-duality/pinned-original-source.tex, source lines 8851–8851. Same-chapter pointer into the author environment remark at source line 8851; invoked at proof line 8895 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G2X: sources/stacks-chow/tags.txt, source lines 20072–20072. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8891 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 0G51: sources/stacks-duality/pinned-original-source.tex, source lines 8721–8721. Same-chapter pointer into the author environment proposition at source line 8721; invoked at proof line 8912 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATX: sources/stacks-duality/pinned-original-source.tex, source lines 4161–4161. Same-chapter pointer into the author environment lemma at source line 4161; invoked at proof line 8917 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ and $g : Y \to Z$ be composable morphisms of FTS_S. With notation as in Remark 0G54 we have $Rg_! \circ Rf_! = R(g \circ f)_!$.*

Proof. By the discussion in Categories, Remark 0G2X it suffices to show that we obtain the same answer if we compute Hom into L in $D^b_{Coh}(\mathcal{O}_Z)$. To do this we compute, using the notation in Remark 0G54, as follows

$$\begin{aligned}
\mathrm{Hom}_Z(Rg_!Rf_!K, L) &= \mathrm{colim}_n \mathrm{Hom}_Z(Rg_!M_n, L) \\
&= \mathrm{colim}_n \mathrm{Hom}_Y(M_n, g^!L) \\
&= \mathrm{Hom}_Y(Rf_!K, g^!L) \\
&= \mathrm{Hom}_X(K, f^!g^!L) \\
&= \mathrm{Hom}_X(K, (g \circ f)^!L) \\
&= \mathrm{Hom}_Z(R(g \circ f)_!K, L)
\end{aligned}$$

The first equality is the definition of $Rg_!Rf_!K$. The second equality is Proposition 0G51 for g. The third equality is the fact that $Rf_!K$ is given by (M_n). The fourth equality is Proposition 0G51 for f. The fifth equality is Lemma 0ATX. The sixth is Proposition 0G51 for $g \circ f$. □